Where Does Abandoned Cropland Go? A Decade of Land-Cover Trajectories in Ohio

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AAG American Association of Geographers

Online Maps & Figures

We ask: **Q1.** Where does abandoned cropland go within ten years? **Q2.** Which pathways does it follow, and where? **Q3.** Do the factors used to predict recultivation also separate these pathways?

Why trajectories?

Abandonment is a process, not an end state, yet large studies treat it as one.

- **A transition, "not a static end state"** (Munroe et al., 2013, p. 471). Fields regrow forest (Cramer et al., 2008), turn to pasture or return to crops (Xie et al., 2024; Crawford et al., 2022), with different payoffs for food, carbon and biodiversity (Zheng et al., 2023; Crawford et al., 2024).
- **Many drivers.** Slope, soil, access and migration shape abandonment (Laue & Arima, 2016; Subedi et al., 2022); Zheng et al. (2023) modeled recultivation with nine factors.
- **Oversimplified at scale.** Large studies map one year's land cover (Xie et al., 2024: in 2020, 53% of 12.3 Mha U.S. abandonment was grassland/pasture), rate ~100 Mha worldwide for two uses only, recultivation or reforestation (Zheng et al., 2023), or sort 2001-2016 U.S. cropland change into four from-to classes (Homer et al., 2020).
- **Now traceable.** A machine-learning abandonment map plus annual 30 m land cover lets us follow each pixel yearly (Fig 1). Start-and-end comparisons miss repeated change: about 40% of changed U.S. land changed more than once in 1985-2016 (Auch et al., 2022).

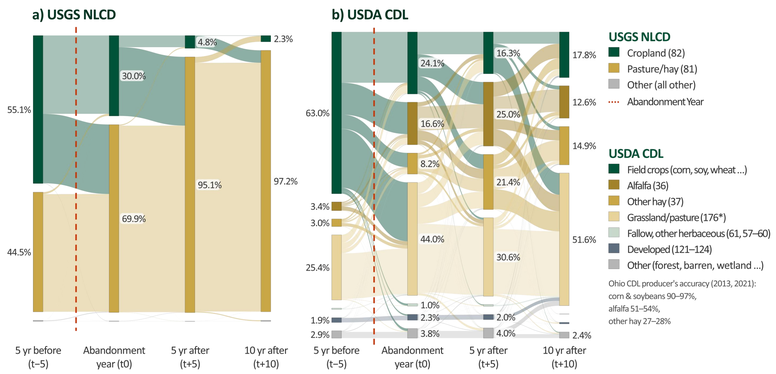


Fig 1. Reading a trajectory: pasture/hay-transition pixels abandoned in 2013 (n = 84,819), followed from t−5 to t+10. NLCD (a) shows one switch to pasture/hay; CDL (b; USDA National Agricultural Statistics Service, 2008-2023) mixes crops, hay and grass.

Data & approach

1. **Abandoned cropland.** Xie et al. (2024), 30 m. Abandoned in 2001-2014. Recultivated and urbanized pixels excluded. Ohio: 1.86 M pixels (≈127,000 ha).
2. **Annual NLCD 1996-2024** (U.S. Geological Survey, 2024), regrouped into four states: Cropland (82), Pasture/hay (81), Developed (21-24), Other; CDL adds crop detail.
3. **Align to the abandonment year (t0)**: t−5 … t+10 (16 years), read as sequences (Liao et al., 2022; Pyrohova et al., 2023). Ohio pixels follow 3,836 distinct sequences.
4. **k-means (k = 5) → five types**: cropland persistence, long-term and short-term pasture/hay transition (pasture/hay before t0 vs. new after t0), urban transition, and other (forest, wetland, water …). The two pasture/hay types are merged for the models.
5. **≈300 m cells** (10 × 10 pixels; ≥10 abandoned pixels; n = 55,784). Each type's share is regressed on ten indicators (Table 1) with a spatial error model (Anselin, 1988).

Table 1. Zheng et al.'s nine factors, rebuilt with U.S. data

Factor	U.S. measure
1 Agro-ecological suitability	Crop calorie yield within 9 km (Ag Census, LAFA)
2 Market accessibility	Distance to major road (OSM)
3 Distance to urban area	Distance to Census urban area (TIGER)
4 Population density (4a, 4b)	Ambient (4a, LandScan) and residential (4b, TIGER)
5 Cropland density	Cropland share within 300 m (CDL)
6 Distance to stable cropland	Distance to land cropped every year (CDL)
7 Abandoned cropland density	Abandoned cropland share within 300 m (Xie et al., 2024)
8 Disaster risk	FEMA risk score
9 Agricultural economic loss	FEMA expected agricultural loss

Q1 Where does it go?

Similar cover, different trajectories: one still farmed, often as alfalfa; one grassed over.

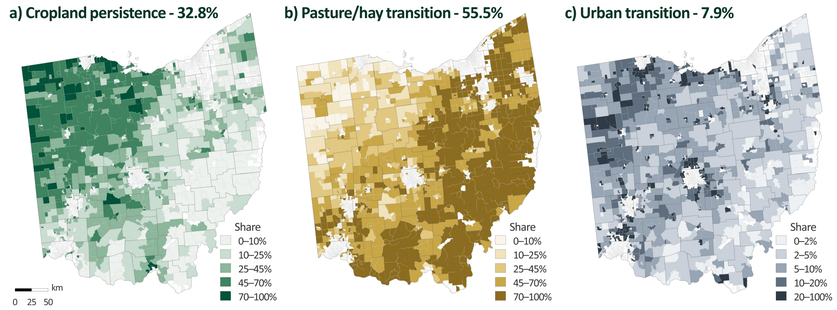


Fig 2. Share of each trajectory type by census tract (tracts with ≥50 abandoned pixels; n = 1,084). Title percentages are Ohio-wide.

- **Pathways (Fig 2):** pasture/hay transition 55.5% (long-term 28.0%, short-term 27.5%; Fig 3), cropland persistence 32.8%, urban 7.9%.
- **A t+10 snapshot differs:** pasture/hay 61.0%, cropland 24.2%, forest 1.5%. In CDL, crops or alfalfa fill 57% of t+1–t+5 on cropland persistence, 30% on pasture/hay, where grass leads (Fig 1b).
- **Versus 11 other Midwest states:** as much herbaceous cover (61.2% vs. 59.5%), but almost none of it grassland/shrub (0.2% vs. 22.7%), and 2.6 times the developed share (10.6% vs. 4.1%).
- **Versus North and Central America:** natural vegetation 2.4% vs. 23% (Potapov et al., 2022).

Q2 Which pathways, where?

Two Ohios: cropland persists in the northwest; pasture/hay held before abandonment fills Appalachia.

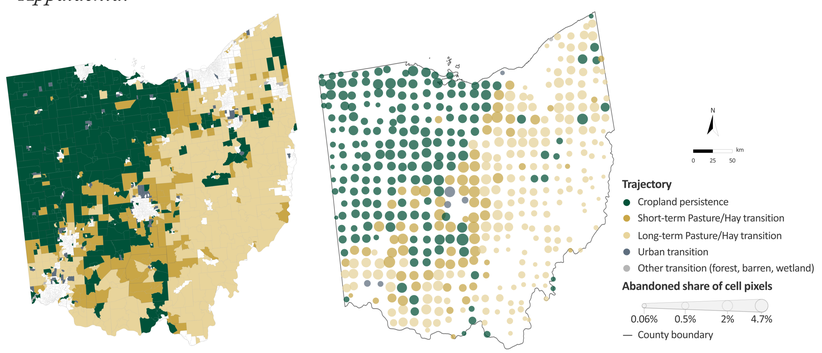


Fig 3. Most common of the five types by census tract (left; n = 1,084) and 15 km cell (right; n = 470, ≥200 abandoned pixels; size = abandoned share of cell pixels).

- **Pasture/hay transition (55.5%).** Long-term (28.0%): NLCD shows pasture/hay before t0 (72% at t−5). Short-term (27.5%): cropland until t0, pasture/hay within five years. 99% are pasture/hay at t+10.
- **Cropland persistence (32.8%).** NLCD still shows cropland at t+5 (98%), but CDL often shows alfalfa instead (23% of t+1–t+5 years).
- **Urban transition (7.9%).** Mostly open-space or low-intensity development (NLCD 21-22: 83%); 44% was already developed at t−5.
- **Where.** Northwestern Ohio: cropland persistence holds 56% of pixels and leads in 80% of tracts. Appalachian Ohio: pasture/hay holds 73% (long-term 47%, short-term 26%); long-term leads in 69% of tracts.

Q3 What separates the pathways?

Partly: nearby cropland and towns separate the pathways; yield barely does.

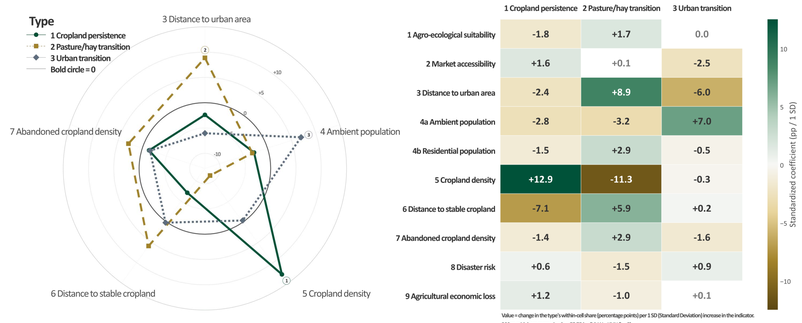


Fig 4. Preliminary results: change in each type's share (percentage points) per 1 SD of each indicator, from spatial error models (n = 55,784 cells). Left: the five strongest. Right: all ten indicators.

- **Cropland persistence** is higher where surrounding cropland is dense (+12.9) and stable cropland is near (−7.1 per SD of distance).
- **Pasture/hay transition** is higher where cropland is sparse (−11.3) and in cells far from urban areas (+8.9) and from stable cropland (+5.9). Nearby abandoned-cropland density is secondary (+2.9, level with residential population density).
- **Urban transition** is higher with larger ambient population (+7.0) and near urban areas (−6.0).
- **Agro-ecological suitability** barely matters (|β| ≤ 1.8), consistent with a review that ranks access and proximity to cities above biophysical drivers (Rey Benayas et al., 2007).

What this adds

1. **Consistent with prior work.** Pasture/hay dominates, as in national data (Xie et al., 2024), and forest is rare, as slow recovery predicts (Crawford et al., 2024).
2. **Beyond prior work.** Zheng's recultivation factors, rebuilt with U.S. data, explain 16-year paths (Zheng et al., 2023); hay or pasture and urban, outside Zheng's two uses, hold 63%.
3. **Combining data.** NLCD gives the long record, CDL the crop detail: alfalfa or other hay on a third of pasture/hay fields at t+5 suggests a shift in land-use intensity (Munroe et al., 2013).
4. **Implication.** Abandonment needs a regional reading: in the northwest, "abandoned" fields are still farmed, often as alfalfa; in Appalachia, they settle into long-held pasture and hay.
5. **Limits.** Definitions differ, as in other U.S. map comparisons (Martin et al., 2025): Xie counts alfalfa as crop, NLCD as hay, and CDL often confuses it with grass. Recultivated fields (Crawford et al., 2022) are excluded.
6. **Next.** Extend to the conterminous U.S. to compare regions, model the three shares jointly, and rebuild cropland indicators without the pixel itself.

References

Anselin, L. (1988). *Spatial econometrics: Methods and models.* Kluwer Academic Publishers.
Auch, R. F., Wellington, D. F., Taylor, J. L., Stehman, S. V., Tollerud, H. J., Brown, J. F., Loveland, T. R., Pengra, B. W., Horton, J. A., Zhu, Z., Midekisa, A. A., Sayler, K. L., Xian, G., Barber, C. P., & Reker, R. R. (2022). Conterminous United States land-cover change (1985-2016): New insights from annual time series. *Land, 11*(2), Article 298.
Cramer, V. A., Hobbs, R. J., & Standish, R. J. (2008). What's new about old fields? Land abandonment and ecosystem assembly. *Trends in Ecology & Evolution, 23*(2), 104-112.
Crawford, C. L., Yin, H., Radeloff, V. C., & Wilcove, D. S. (2022). Rural land abandonment is too ephemeral to provide major benefits for biodiversity and climate. *Science Advances, 8*(21), Article eabm8999.
Crawford, C. L., Wiebe, R. A., Yin, H., Radeloff, V. C., & Wilcove, D. S. (2024). Biodiversity consequences of cropland abandonment. *Nature Sustainability, 7*(12), 1596-1607.
Homer, C., Dewitz, J., Jin, S., Xian, G., Costello, C., Danielson, P., Gass, L., Funk, M., Wickham, J., Stehman, S., Auch, R., & Riitters, K. (2020). Conterminous United States land cover change patterns 2001-2016 from the 2016 National Land Cover Database. *ISPRS Journal of Photogrammetry and Remote Sensing, 162*, 184-199.
Laue, J. E., & Arima, E. Y. (2016). Spatially explicit models of land abandonment in the Amazon. *Journal of Land Use Science, 11*(1), 48-75.
Liao, T. F., Bolano, D., Brzinsky-Fay, C., Cornwell, B., Fasang, A. E., Helske, S., Piccarreta, R., Raab, M., Ritschard, G., Struffolino, E., & Studer, M. (2022). Sequence analysis: Its past, present, and future. *Social Science Research, 107*, Article 102772.
Martin, G., Austin, K., Lark, T., Lee, S., & Clark, C. M. (2025). Tracking cropland transitions: A comparative analysis of U.S. land cover change data. *PLOS ONE, 20*(3), Article e0313880.
Munroe, D. K., van Berkel, D. B., Verburg, P. H., & Olson, J. L. (2013). Alternative trajectories of land abandonment: Causes, consequences and research challenges. *Current Opinion in Environmental Sustainability, 5*(5), p. 471-476.
Potapov, P., Turubanova, S., Hansen, M. C., Tyukavina, A., Zalles, V., Khan, A., Song, X.-P., Pickens, A., Shen, Q., & Cortez, J. (2022). Global maps of cropland extent and change show accelerated cropland expansion in the twenty-first century. *Nature Food, 3*(1), 19-28.
Pyrohova, S., Hu, J., & Corcoran, J. (2023). Urban land use transitions: Examining change over 19 years using sequence analysis. The case of South-East Queensland, Australia. *Environment and Planning B: Urban Analytics and City Science, 50*(9), 2579-2593.
Rey Benayas, J. M., Martins, A., Nicolau, J. M., & Schulz, J. J. (2007). Abandonment of agricultural land: An overview of drivers and consequences. *CAB Reviews: Perspectives in Agriculture, Veterinary Science, Nutrition and Natural Resources, 2*(057), 1-14.
Subedi, Y. R., Kristiansen, P., & Cacho, O. (2022). Drivers and consequences of agricultural land abandonment and its reutilisation pathways: A systematic review. *Environmental Development, 42*, Article 100681.
National Agricultural Statistics Service. (2008–2023). Cropland Data Layer [Data set]. U.S. Department of Agriculture. Retrieved September 2, 2026
U.S. Geological Survey. (2024). Annual NLCD Collection 1 science products (Version 1.1) [Data set].
Xie, Y., Spawn-Lee, S. A., Radeloff, V. C., Yin, H., Robertson, G. P., & Lark, T. J. (2024). Cropland abandonment between 1986 and 2018 across the United States: Spatiotemporal patterns and current land uses. *Environmental Research Letters, 19*(4), Article 044009.
Zheng, Q., Ha, T., Prishchepov, A. V., Zeng, Y., Yin, H., & Koh, L. P. (2023). The neglected role of abandoned cropland in supporting both food security and climate change mitigation. *Nature Communications, 14*(1), Article 6083.